

National University of Computer and Emerging Sciences

Applied Machine Learning CS(4104)

Date: Nov 7th 2025

Course Instructor(s)

Mr. Minhaz Raza

Sessional-II Exam

Total Time: 1 Hours

Total Marks: 15

Total Questions: 04

Semester: Fall-2025

Campus: Karachi

Dept: SE

Roll No

Section

Student Signature

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Attempt all the questions.

CL02: Formulate and execute solutions to the machine learning problems

Question 1: A software engineering team is using a classifier to predict whether a module will be buggy or non-buggy before release. Given $P(C_1) = 0.25$, $P(C_2) = 0.75$, $P(x|C_1) = 0.8$, $P(x|C_2) = 0.3$ [2.5 Marks]

		Predicted Class		Total
		Buggy	Not Buggy	
Actual Class	Buggy	40	10	50
	Not Buggy	20	130	150
	Total	60	140	200

a. Calculate Bayes Error Rate.

$$P(C_1) = 50/200 = 0.25$$

$$P(C_2) = 150/200 = 0.75$$

Classifier's conditional probabilities:

$$P(x | C_1) = 0.8$$

$$P(x | C_2) = 0.3$$

Step 1: Compute the *posterior probabilities*

We use Bayes' theorem for a sample xxx (a module predicted buggy):

$$P(C_1|x) = \frac{0.8(0.25)}{0.8(0.25) + 0.3(0.75)} = \frac{0.2}{0.2 + 0.225} = 0.47$$

$$P(C_2|x) = 1 - 0.47 = 0.53$$

Step 2: Compute Bayes Error

Bayes classifier picks the class with **maximum posterior probability**, so error = probability of the other class:

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$$\text{Bayes Error} = \min(P(C1|x), P(C2|x))$$

$$\text{Bayes Error} = 0.47$$

Bayes Error Rate = 0.47 (47%)

b. If you increase $P(C_1)$ to 0.7, does Bayes error increase, decrease, or remain the same? Explain.

When the prior probability $P(C_1)$ (buggy modules) increases, the classifier is more confident that a random module is buggy. Because the posterior for C_1 rises sharply ($0.47 \rightarrow 0.86$), the decision becomes less ambiguous. Therefore, the **Bayes error decreases** — the classifier is making fewer uncertain or wrong decisions on average.

c. You observe that the 1-NN classifier has an error rate of 0.32 on a large dataset. Using the bound: $P_{1NN} \leq 2P^*(1-P^*)$ estimate the possible range of Bayes Error P^* . (Hint: You'll need to reason backward — this question checks algebraic and conceptual insight.)

We need values of P^* satisfying:

$$2P^*(1 - P^*) \geq 0.32.$$

Solve equation $2P(1 - P) = 0.32$. Divide by 2:

$$P - P^2 = 0.16 \Rightarrow P^2 - P + 0.16 = 0.$$

Compute discriminant $D = 1 - 4 \times 0.16 = 1 - 0.64 = 0.36$. $\sqrt{D} = 0.6$. Roots:

$$P = \frac{1 \pm 0.6}{2} \Rightarrow P_1 = 0.20, P_2 = 0.80.$$

The quadratic is concave-down between roots, so the inequality holds for $P \in [0.20, 0.80]$. But Bayes error P^* cannot exceed 0.5 (because $\min(p_1, p_2) \leq 0.5$ always), so intersect with $[0, 0.5]$:

Possible Bayes Error range = $[0.20, 0.50]$ (i.e. between 20% and 50%).

CL03 : Understand the core theoretical concepts serve as foundations of ML algorithms

Question 2: A project manager wants to use a decision tree to predict whether a software release will be delayed based on three attributes: Test Coverage, Team Skill, and Requirement Change. [5 Marks]

Release	Test Coverage	Team Skill	Requirement Change	Delayed
R1	High	High	Low	No
R2	Medium	Medium	Medium	Yes
R3	Low	Low	High	Yes
R4	High	Medium	Low	No
R5	Low	Medium	High	Yes
R6	Medium	High	Medium	No
R7	Medium	Low	High	Yes
R8	High	High	Medium	No

- Calculate entropy for target variable "Delayed."
- Compute information gain for each attribute (Test Coverage, Team Skill, Requirement Change).
- Identify the best attribute for the root node.
- Show one subtree expansion with the second-best split.

1) Entropy of target

$$\text{Entropy}(S) = 1.000000$$

2) Information gain for each attribute

All three attributes (Test Coverage, Team Skill, Requirement Change) split the data into groups with the same group distributions in this dataset — they produce equal IG. I compute one explicitly (others are symmetric):

Test Coverage (High, Medium, Low)

- High: 3 releases → (Yes=0, No=3) → entropy = 0.000000
- Medium: 3 releases → (Yes=2, No=1) → entropy = 0.9182958341
- Low: 2 releases → (Yes=2, No=0) → entropy = 0.000000

Weighted entropy:

$$\frac{3}{8} \cdot 0 + \frac{3}{8} \cdot 0.9182958341 + \frac{2}{8} \cdot 0 = 0.3443609378$$

Information Gain:

$$\text{IG}(S, \text{Test Coverage}) = 1.000000 - 0.3443609378 = 0.6556390622$$

(= 0.65564)

Team Skill and Requirement Change

Calculated the same way — each yields IG = 0.6556390622 (tie).

3) Best attribute for root

There is a **three-way tie**: Test Coverage, Team Skill, and Requirement Change all give IG = 0.65564. Any of them is a valid root: pick based on domain knowledge or tie-break rule.

4) Example subtree expansion (choose Test Coverage as root)

Root split by Test Coverage:

- High → releases: R1 (No), R4 (No), R8 (No) → all No → leaf (entropy 0)
- Low → releases: R3 (Yes), R5 (Yes) → all Yes → leaf (entropy 0)
- Medium → releases: R2 (Yes), R6 (No), R7 (Yes) → mixed → entropy = 0.9182958341

Now expand the Medium branch. Consider splitting Medium by Requirement Change:

Medium branch members and their Requirement Change:

- R2 → Medium change → Delayed = Yes
- R6 → Medium change → Delayed = No
- R7 → High change → Delayed = Yes

Split by Requirement Change inside Medium branch:

- Medium change subgroup: R2 (Yes), R6 (No) → entropy = 1.000000
- High change subgroup: R7 (Yes) → entropy = 0.000000

Weighted entropy (for this branch):

$$\frac{2}{3} \cdot 1 + \frac{1}{3} \cdot 0 = 0.6666666667$$

Information gain inside the Medium branch:

$$\text{Gain} = 0.9182958341 - 0.6666666667 = 0.2516291674$$

(= 0.25163) — so splitting that Medium branch by Requirement Change reduces uncertainty.

CLO3 : Understand the core theoretical concepts serve as foundations of ML algorithms

Question 3: a) A logistic regression model for predicting whether a software module contains a defect is: $\text{logit}(p) = -4 + 0.2X_1 + 0.4X_2$ where $X_1 = 10$ (Number of Change Requests), $X_2 = 8$ (Communication Gaps Detected). Now, if X_1 increases by 5, but X_2 decreases by 2, find new probability and compare it to the old one and explain the intuition. ($p > 0.5$ for it's defectiveness)

(a) Model $\text{logit}(p) = -4 + 0.2X_1 + 0.4X_2$

Initial: $X_1 = 10$, $X_2 = 8$.

1. Compute z_0 :

$$z_0 = -4 + 0.2(10) + 0.4(8) = -4 + 2 + 3.2 = 1.2.$$

2. Use sigmoid: compute $e^{-z_0} = e^{-1.2}$.

We know $e^{1.2} \approx 3.32011692$, so $e^{-1.2} = 1/3.32011692 \approx 0.30119421$.

$$p_0 = \frac{1}{1 + e^{-1.2}} = \frac{1}{1 + 0.30119421} = \frac{1}{1.30119421} \approx 0.76852478.$$

After change: $X'_1 = 15$, $X'_2 = 6$.

3. Compute z_1 :

$$z_1 = -4 + 0.2(15) + 0.4(6) = -4 + 3 + 2.4 = 1.4.$$

4. Sigmoid: $e^{-z_1} = e^{-1.4}$.

$e^{1.4} \approx 4.054054$ so $e^{-1.4} = 1/4.054054 \approx 0.24659696$.

$$p_1 = \frac{1}{1 + e^{-1.4}} = \frac{1}{1 + 0.24659696} = \frac{1}{1.24659696} \approx 0.80218307.$$

Summary (a): p increases from ≈ 0.7685 to ≈ 0.8022 . Both > 0.5 , so predicted defective both times. The odds ratio corresponding to the net z -change of 0.2 is $\exp(0.2) \approx 1.22140$ ($\approx 22.14\%$ increase in odds).

b) A defect prediction model is given by: $\text{logit}(p) = -3 + 0.4X_1 + 0.9X_2$. For a module with $X_1=5$, $X_2=2$, compute p . Then, if X_2 increases by 1, calculate the change in odds, not in probability. (Hint: Odds = $p / (1 - p)$; use ratio of new odds to old odds.) and interpret your results and findings

(b) Model $\text{logit}(p) = -3 + 0.4X_1 + 0.9X_2$

Given $X_1 = 5$, $X_2 = 2$.

1. Compute z_0 :

$$z_0 = -3 + 0.4(5) + 0.9(2) = -3 + 2 + 1.8 = 0.8.$$

2. Sigmoid: $e^{-0.8} \approx 0.44932896$.

$$p_0 = \frac{1}{1 + e^{-0.8}} = \frac{1}{1 + 0.44932896} = \frac{1}{1.44932896} \approx 0.68997448.$$

If X_2 increases by 1 (so $X'_2 = 3$), z increases by 0.9 to $z_1 = 1.7$.

3. $e^{-1.7} \approx 0.18268352$.

$$p_1 = \frac{1}{1 + e^{-1.7}} = \frac{1}{1 + 0.18268352} = \frac{1}{1.18268352} \approx 0.84553473.$$

Change in odds (useful interpretation):

- Initial odds = $\frac{p_0}{1 - p_0} = \frac{0.68997448}{0.31002552} \approx 2.22554093$ (which equals $e^{0.8}$).
- New odds = $\frac{p_1}{1 - p_1} = \frac{0.84553473}{0.15446527} \approx 5.47394739$ (which equals $e^{1.7}$).

- Odds ratio = $\frac{5.47394739}{2.22554093} \approx 2.460$ which equals $\exp(0.9)$. So the odds are multiplied by ≈ 2.46 ($\approx 146\%$ increase).

Summary (b): starting $p \approx 0.68997$. Increasing X_2 by 1 makes $p \approx 0.84553$. The odds are multiplied by $\exp(0.9) \approx 2.46$.

CLO3 : Understand the core theoretical concepts serve as foundations of ML algorithms

- Question 4:** Training set examples $(x_1, y_1) \dots (x_4, y_4)$ with labels $y = [+1, +1, +1, -1]$. Initial weights $\hat{w}_i^{(1)} = 0.25$ each. Weak learner h_1 misclassified's only x_4 and correctly classifies the others three. [5 Marks]
- Compute the error ϵ_1 of h_1
 - Compute the classifier weight $\alpha_1 = 1/2 * \ln(1-\epsilon_1)/\epsilon_1$
 - Compute updated (unnormalized) weights and then the normalized weights $\hat{w}_i^{(2)}$.
 - Given three weak classifiers with outputs on a sample $h_1(x)=+1, h_2(x)=-1, h_3(x)=+1$ and their alphas $\alpha_1=0.6931, \alpha_2=0.4236, \alpha_3=0.2027$. What is the AdaBoost ensemble prediction $H(x)$?

Solution (step-by-step)

a) ϵ_1 = sum of weights of misclassified examples = weight of x_4 = 0.25.

$$b) \alpha_1 = \frac{1}{2} \ln \frac{1-\epsilon_1}{\epsilon_1} = \frac{1}{2} \ln \frac{0.75}{0.25} = \frac{1}{2} \ln 3.$$

Numeric: $\ln 3 \approx 1.098612 \Rightarrow \alpha_1 \approx 0.549306$.

Useful identities: $e^{\alpha_1} = \sqrt{3} \approx 1.73205$, $e^{-\alpha_1} \approx 0.57735$.

c) Weight update rule (unnormalized):

- If example correctly classified: $w_i' = w_i \cdot e^{-\alpha_1}$.
- If misclassified: $w_i' = w_i \cdot e^{\alpha_1}$.

Compute:

- For each of the three correct examples: $0.25 \times e^{-\alpha_1} = 0.25 \times 0.57735 \approx 0.144338$.
- For the misclassified example (x_4): $0.25 \times e^{\alpha_1} = 0.25 \times 1.73205 \approx 0.433013$.

Normalization constant $Z = 3 \times 0.144338 + 0.433013 \approx 0.866027$.

Normalized weights:

- Correct examples: $0.144338/0.866027 \approx 0.1666667$ (i.e. $1/6$).
- Misclassified example: $0.433013/0.866027 \approx 0.5$.

So $w^{(2)} = [1/6, 1/6, 1/6, 1/2]$.

Compute weighted sum: $S = \alpha_1(+1) + \alpha_2(-1) + \alpha_3(+1) = 0.6931 - 0.4236 + 0.2027 = 0.4722$
(approx). $\text{sign}(S) = +1$. So ensemble predicts +1.